



POLITECNICO
MILANO 1863

Hedging Competition

Group 10

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1 Introduction

This project is all about stepping into the shoes of a market maker and testing our skills in calibration, pricing, and hedging. We worked with a dataset of SP 500 options from mid-2017, built models to capture the volatility surface, and then used those models to price and hedge a structured product called the Miami Crocodile. In the end, the goal is simple: design a hedging approach that keeps the portfolio stable and minimizes risk.

2 Calibration of Discount Factor and Forward Rates

The option dataset was first extracted and processed by applying filtering criteria to remove instruments not suitable for calibration. In particular, penny options, contracts with excessively wide bid-ask spreads, and options with marginal deltas were discarded. The cleaned data were then organized into a cell array structure, where the i -th cell contains the set of strikes, call prices, and put prices available at the i -th observable market date.

The discount factors and forward prices were calibrated using the classical put-call parity relation:

$$C_{t_0}(T, K) - P_{t_0}(T, K) = B(t_0, T) \cdot (F(t_0, T) - K) \quad (1)$$

where $C_{t_0}(T, K)$ and $P_{t_0}(T, K)$ denote the call and put prices, K is the strike, $B(t_0, T)$ is the discount factor, and $F(t_0, T)$ is the forward price.

For a fixed maturity T , the discount factor $B(t_0, T)$ corresponds to the slope of the line implied by the above relation as a function of the strike K , while the forward price $F(t_0, T)$ is identified with the intercept divided by the discount. This procedure allowed us to consistently calibrate both discount factors and forward prices across maturities using the available option quotes.

3 Calibration of the Volatility Surface

We calibrated two alternative models for the equity volatility surface: the Normal Inverse Gaussian (NIG) model and the Power Scaling ATS model. Both models were fitted by minimizing the squared error between market option prices and model prices as a function of the parameters, using out-of-the-money calls and puts as calibration instruments.

In the calibration of the ATS model, we imposed constraints that ensure the resulting log-return process is additive.

Once the optimal parameter vectors were obtained for each model, we compared the implied volatilities generated by the models with the market implied volatilities. As expected,

the ATS model performed significantly better, especially for shorter maturities. This superior performance motivated the choice of the ATS model as the basis for our hedging strategy.

We report the graphs of the implied volatilities for three maturities, comparing the NIG and ATS models with the market surface:

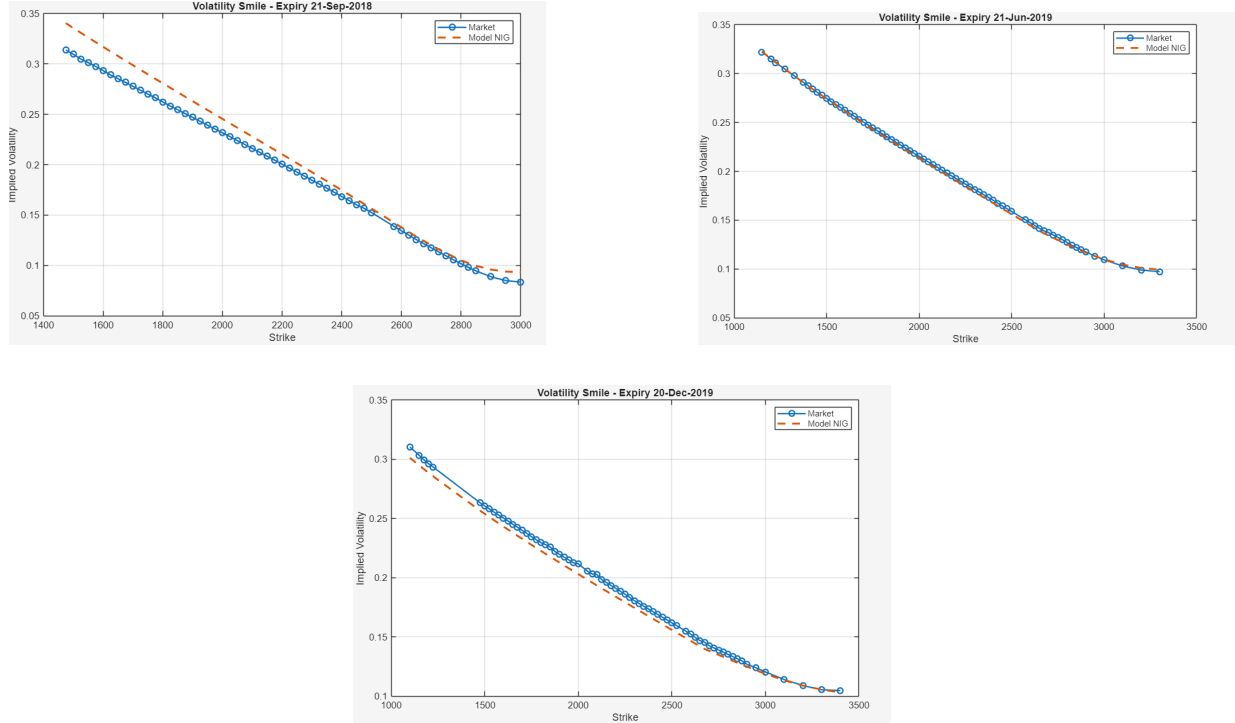


Figure 1: NIG model

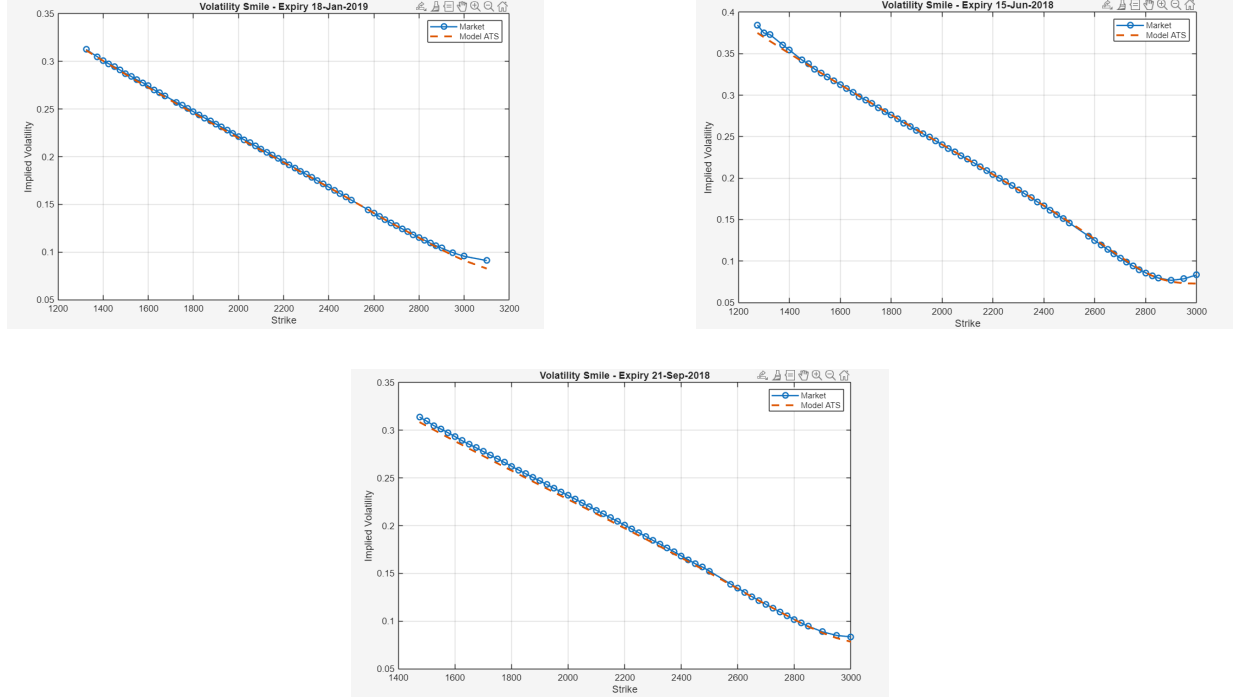


Figure 2: Power Scaling ATS model

Model	RMSE	MAPE
NIG	0.0334	9.44%
ATS	0.0059	2.20%

Table 1: Calibration error metrics for NIG and ATS models.

As can be seen from these error metrics, the ATS model is much more accurate than the NIG model across the entire volatility surface.

4 Simulation Algorithm

The chosen ATS Power Scaling model is based on the structure of the log-returns of the forward price. Once the model parameters are calibrated, we have access to its characteristic function, which fully describes the distribution of the log-returns.

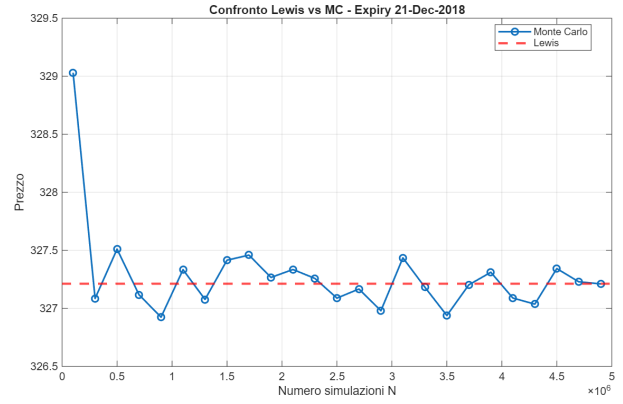
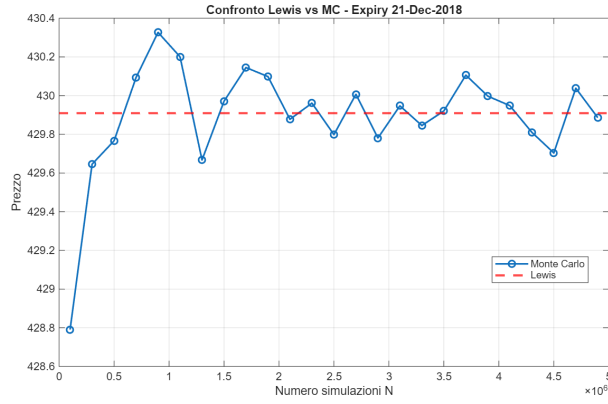
The simulation algorithm therefore relies on generating realizations of the log-returns f_t through the characteristic function of the model. This allows us to construct simulated paths of the forward price under the ATS dynamics.

To verify the correctness of the simulation procedure, we priced European call options using two approaches:

- The “exact” pricing formula derived by Lewis, which exploits the characteristic function of the model.

- A Monte Carlo simulation of the log-returns, where the option price is obtained as the discounted expected payoff.

By increasing the number of Monte Carlo simulations, we observed that the simulated call prices converge to the analytical Lewis formula, as shown in the following graphs.



5 Pricing the Miami Crocodile Certificate

The certificate to be priced includes the possibility of automatic early settlement (autocall) at predetermined observation dates. In addition, the final payoff, in the absence of autocall, depends on the underlying price observed at those dates.

This structure requires simulating log-returns of the forward price along consistent trajectories, since the payoff depends on the joint evolution of the underlying across multiple dates.

In the ATS Power Scaling model, the process f_t is an additive process. This implies that the increments of f_t are independent, therefore the characteristic function of each increment can be obtained as ratio between the characteristic functions of the log-return evaluated at the two time points.

Therefore, the simulation procedure is based on generating:

$$f_1, \quad f_2 - f_1, \quad f_3 - f_2, \quad \dots$$

so that we obtain values of f_t consistent with a single trajectory of the forward price. This ensures that the simulated paths correctly capture the dependence structure required to evaluate both autocall events and the final payoff of the certificate.

Once the simulations of the log-returns were obtained, we checked the underlying prices at the observation dates, constructed the vector of discounted payoffs for the different cases, and finally computed the average across all simulated trajectories.

The resulting normalized price of the Miami Crocodile certificate is:

$$\text{Normalized Price} = 1.00487 \text{ USD}$$

with a 99% confidence interval given by:

$$[1.00460 \text{ USD}, 1.00514 \text{ USD}] .$$

6 Automatic Hedging Strategy

The hedging strategy was designed to neutralize some of the main risk factors of the certificate, namely delta and vega.

We first computed the delta of the certificate numerically using a central difference scheme, applying positive and negative bumps to the initial forward price. This allowed us to approximate the sensitivity of the certificate value with respect to movements in the underlying.

Subsequently, we bumped the entire volatility surface, recalibrated the model with the option prices obtained under the bumped volatilities, and calculated the numerical vega of the certificate. Observing that the certificate exhibited significant sensitivity to volatility changes, we opted to hedge vega in addition to delta.

The instrument chosen to hedge vega was the at-the-money straddle, which provides vega equal to twice the vega of a single call option while maintaining a small delta exposure. To avoid introducing an excessive number of products into the portfolio, and given that the first market-observable dates are far from the certificate's observation dates, we decided to hedge vega using straddles only at five selected market dates. These dates absorb the majority of the certificate's vega exposure.

To determine the price movement of the certificate to be hedged with a single straddle, we bumped the entire volatility smile corresponding to the chosen date, recalibrated the model, and measured the resulting variation in the certificate price. This procedure ensured that the straddle hedge was aligned with the volatility risk profile of the certificate.

Therefore, the automatic hedging strategy is implemented as a loop over the set of future dates:

1. Extract market data.
2. Calibrate the model.
3. Price the certificate.
4. Compute local vegas at the dates where straddle positions are opened.
5. Compute the delta of the portfolio (certificate + straddles) and open a spot position to neutralize the final portfolio delta.
6. Compute the mark-to-market (MtM) of the hedged portfolio, which is stored for the next iteration.
7. In the following iteration, reprice the previously opened positions with updated market data, thereby tracking the daily P&L.

6.1 Comments on Hedging Performance

As expected, the delta of the certificate is positive, since a higher spot level increases the probability of reaching the trigger value while reducing the likelihood of hitting the lower barrier. Conversely, the vega is negative, because more volatile trajectories of the underlying raise the probability of touching the lower barrier. Consequently, the hedging positions opened are short straddles and long spot (since we are short the certificate).

Running the strategy on past dates up to the present, processed market data showed that some options purchased the previous day were no longer available. In these cases, we approximated the price of the missing product using the updated volatility surface, interpolating at the correct strike and applying the Black formula (`blkprice`). This scenario produced the largest shocks in the P&L, reaching a maximum of 40K.

On most trading days, however, the variation in the certificate price was more pronounced than the daily P&L recorded, which supports the effectiveness of the hedging strategy in stabilizing portfolio performance.

The execution time of the automatic hedging function is not negligible when a large set of dates is provided, since for each day the code performs six calibrations: one initial calibration and five additional calibrations to compute the vegas required for hedging with the five straddle positions.

7 Backtesting and Performance Evaluation

We performed a backtesting exercise using the three future dates provided. For each date, we computed both the P&L of the hedged portfolio and the variation that the certificate would have experienced without hedging.

The results of the hedged portfolio P&L are:

−17357 USD, −40262 USD, −69 USD

These values highlight the stabilizing effect of the hedging strategy compared to the unhedged certificate.

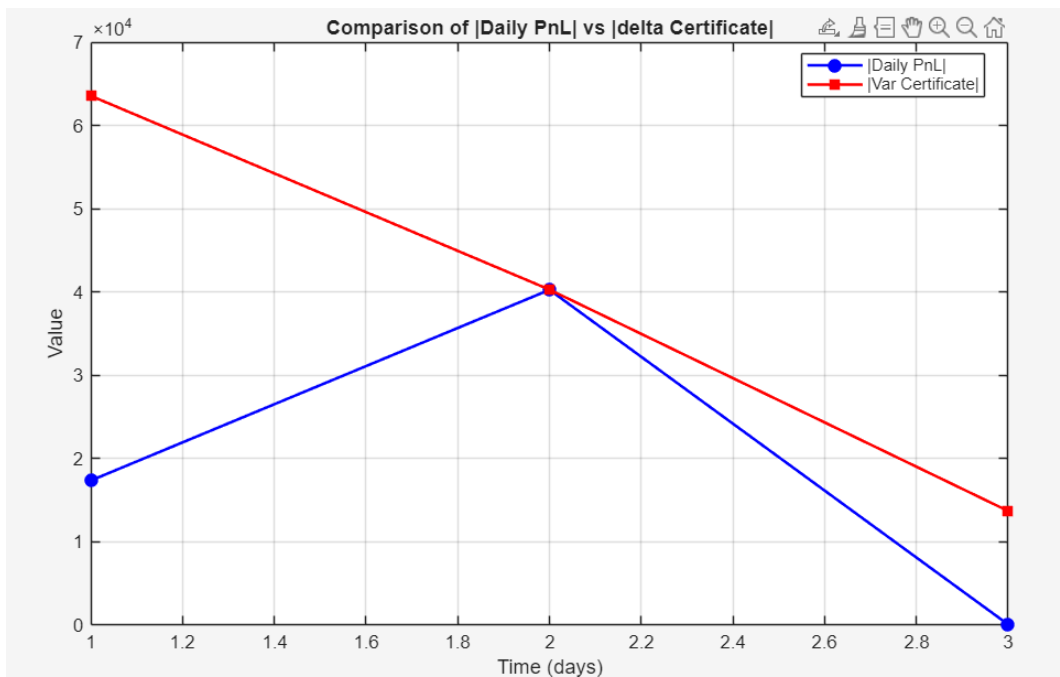


Figure 4: Comparison between the P&L of the hedged portfolio and the unhedged certificate.

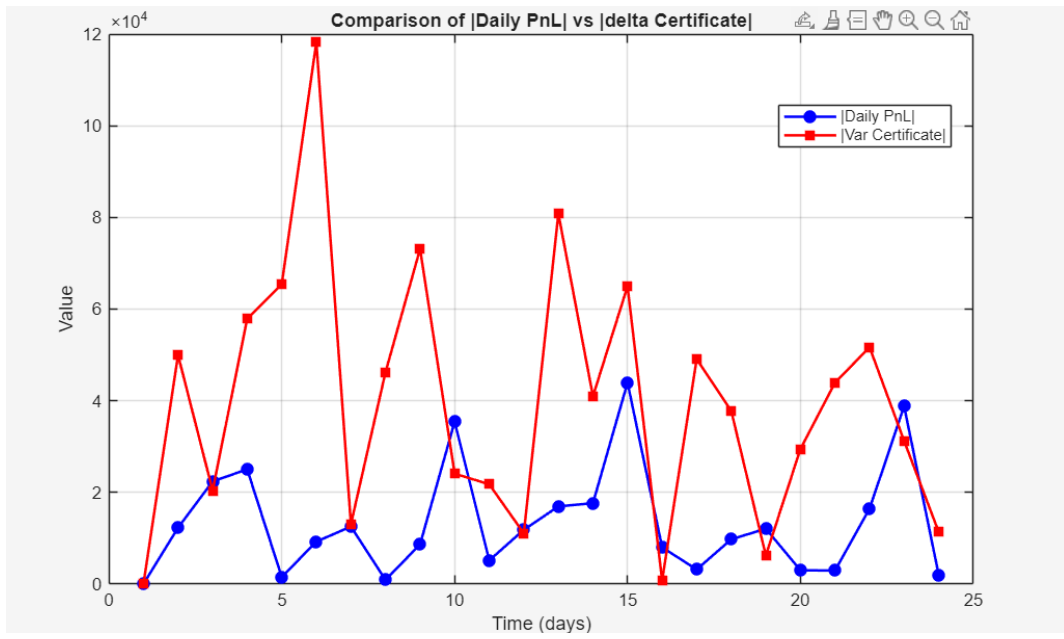


Figure 5: Comparison between the P&L of the hedged portfolio and the unhedged certificate with start date 2017/11/09.